

Algebraic geometry 2026

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Chapter 1

Mynotes from 2026 class

Some references used here:

- Mumford's red book [2]
- Humphreys Linear algebraic groups [1]

1.1 Lecture 1 and 2

First some definitions and lemmas worth to remember.

- (Integral extension). Let R be a ring and $S \subset R$ a subring. We say that R is integrally dependent over S if every $r \in R$ is integral over S , that is, there exist a monic polynomial $f \in S[X]$ such that $f(r) = 0$.

Lemma 1.1.1. Let $S \subset R$ be rings. A finite subset of elements $b_1, \dots, b_m \in R$ are all integral over S if and only if $S[b_1, \dots, b_m]$ is finitely generated (as a S -module) over S .

Proof: ($\Rightarrow$) Let $b \in R$ be integral over S , i.e., there exist

$$f = X^d + a_1 X^{d-1} + \dots + a_d \in S[X] \quad (1.1)$$

such that $f(b) = 0$. Then, using long division, we see that $S[b] = S + \dots + Sb^{d-1}$ which is finitely generated. Now, if $b_1, \dots, b_m$ are all integral over S , we can argue by induction that $S[b_1, \dots, b_{m-1}]$ is finitely generated over S . However, since b_m is integral over $S[b_1, \dots, b_{m-1}]$, $S[b_1, \dots, b_m]$ is finitely generated over $[b_1, \dots, b_{m-1}]$ and it must be finitely generated over S as well.

($\Leftarrow$) Suppose that $S[b_1, \dots, b_m] = S\beta_1 + \dots + S\beta_n$ for some $\beta_i \in R$. Then for any $b \in R$ which fixes $S[b_1, \dots, b_m]$, we have $b\beta_i = \sum_j a_{ij}\beta_j$ for some $(a_{ij}) \in \text{Mat}_n(S)$. Therefore

$$0 = (bI - (a_{ij}))^*(bI - (a_{ij}))v = \det(bI - (a_{ij}))v \quad (1.2)$$

where $v = (\beta_1, \dots, \beta_n)^T$. Therefore $\det(bI - (a_{ij}))\beta_i = 0, \forall i$ and since $1 \in S\beta_1 + \dots + S\beta_n$, we see that $\det(bI - (a_{ij})) = 0$. Since $\det(X - (a_{ij})) \in S[X]$ is monic, we are done. $\square$

Corollary 1.1.2. (Transitive property of integral dependence). Given rings $R \supset S \supset T$, if R is integral over S and S is integral over T , then R is integral over T .

Question: Do we need Noetherian hypothesis using this statement?

Proof: Let $r \in R$, then by hypothesis there exist $f = X^d + s_1X^{d-1} + \dots \in S[X]$ such that $f(r) = 0$. Define $A = T[s_1, \dots, s_d]$, then $A[r]$ is a finitely generated A -module because $S \subset A$ and lemma (1.1.1). Now, since A is a finitely generated T -module by lemma (1.1.1), we see that $A[r]$ is a finitely generated T -module. Now, **assuming T is Noetherian** we see that $T[r]$ is finitely generated as a T -module. $\square$

Remark 1.1.3. We can certainly say (using the previous proof) that if $r \in R$, then $S[r]$ is finitely generated as a T -module.

Lemma 1.1.4. Let $K \supset k$ (k a perfect field) be a finitely generated extension field with transcendence degree $\partial_k(K) = d$, then there exist generators $z_1, \dots, z_d, z_{d+1}$ such that

- (1) $K = k(z_1, \dots, z_{d+1})$
- (2) $\{z_1, \dots, z_d\}$ are algebraically independent
- (3) z_{d+1} is separable over $k(z_1, \dots, z_d)$

Lemma 1.1.5. (Noether's normalization lemma - Mumford's red book). Let R be a finitely generated algebra over k . Assume also that R is an integral domain. If R has transcendence degree n over k (which means that $\partial_k \text{Frac}(R) = n$), then there exist $x_1, \dots, x_n \in R$ algebraically independent and such that R is integral over $k[x_1, \dots, x_n]$.

TODO: Study prove this version of the lemma using the statement made in class.

Lemma 1.1.6. (Noether's normalization lemma - Statement made in class). Let R be a finitely generated k -algebra. Then there exist $x_1, \dots, x_n \in R$ algebraically independent over k such that R is integral over $k[x_1, \dots, x_n]$.

Proof (Nagata): Since R is finitely generated as a k -algebra, there exist generators $y_1, \dots, y_m$ such that $R = k[y_1, \dots, y_m]$. Now suppose that $y_1, \dots, y_m$ are algebraically independent, then we can take $x_i = y_i$. If not, then there exist $p \in k[Y_1, \dots, Y_m] \setminus 0$ such that $p(y_1, \dots, y_m) = 0$. Write

$$p(Y_1, \dots, Y_m) = \sum_{\alpha \in \mathbb{Z}_{\geq 0}^m} a_\alpha Y_1^{\alpha_1} \dots Y_m^{\alpha_m} \quad (1.3)$$

and define a new polynomial

$$\begin{aligned} q(Y_1, \dots, Y_m) &= p(Y_1, Y_2 + Y_1^{r_2}, \dots, Y_m + Y_1^{r_m}) \\ &= \sum_{\alpha} a_{\alpha} (Y_1^{\alpha_1 + r_2 \alpha_2 + \dots + r_m \alpha_m} + \dots) \end{aligned} \quad (1.4)$$

for some choice of positive integers $r_2, \dots, r_m$. Take a positive number $b > \alpha_i, \forall \alpha_i$ such that $a_{\alpha} \neq 0$. If we take $r_i = b^i$, then $\alpha_1 + \alpha_2 b + \dots + \alpha_m b^{m-1}$ are all different numbers in base b . So, from different powers of the form $\alpha_1 + r_2 \alpha_2 + \dots + r_m \alpha_m$ we have a maximal one $d = \max\{\alpha_1 + \dots + r_m \alpha_m : a_{\alpha} \neq 0\}$ and

$$q(Y_1, \dots, Y_m) = aY_1^d + \text{smaller degree terms} \in k[Y_2, \dots, Y_m][Y_1], \quad a \neq 0. \quad (1.5)$$

Now, if we let $z_i = y_i - y_1^{r_i} \in R, i = 2, \dots, m$, we see that $q(y_1, z_2, \dots, z_m) = p(y_1, \dots, y_m) = 0$ and y_1 is integral over $k[z_2, \dots, z_m]$. By induction on number of generators, there exist $x_1, \dots, x_n$ algebraically independent such that $k[z_2, \dots, z_m]$ is integral over $k[x_1, \dots, x_n]$. So, by lemma 1.1.1 and its corollary (1.1.1), we see that $k[y_1, z_2, \dots, z_m]$ is integral over $k[x_1, \dots, x_m]$. Now, by definition, it's clear that $k[y_1, z_2, \dots, z_m]$ is a finitely generated R -module. So by the corollary (1.1.2) again, we see that R is integral over $k[x_1, \dots, x_m]$. $\square$

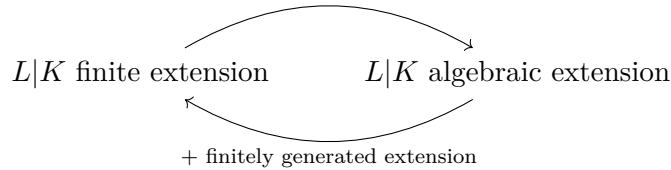
Lemma 1.1.7. Let $S \subset R$ be a ring extension. If R is a field, which integral over S , then S is a field.

Proof: Let $s \in S \setminus 0$ and $r \in R$ be its inverse. Since R is integral over S , there exist $f = X^d + a_1 X^{d-1} + \dots \in S[X]$ be such that $f(r) = 0$. Therefore

$$0 = s^{d-1} f(r) = r + a_1 + sa_2 + \dots + s^{d-1} a_0 \Rightarrow r \in S. \quad (1.6)$$

$\square$

Remark 1.1.8. Let $L|K$ be a field extension. Then



Theorem 1.1.9. (Zariski). Let k be a field and $\mathfrak{m} \subset R = k[X_1, \dots, X_m]$ be a maximal ideal. Then $L = R/\mathfrak{m}$ is a finite extension field of k .

Proof: By Noether normalization lemma (1.1.6), there exist $a_1, \dots, a_n \in L$ algebraically independent such that L is integral over $k[a_1, \dots, a_n]$. By lemma (1.1.7), we conclude

that $k[a_1, \dots, a_n]$ is a field. Suppose that $n > 0$, so a_1 has an inverse. This means that $a_1 g(a_1, \dots, a_n) = 1$ for some $g \in k[X_1, \dots, X_n]$, which contradicts the fact that $a_1, \dots, a_n$ are algebraically independent. Then we conclude that L is an integral (and then algebraic) over k . However, since L is a finitely generated extension ($L = k(\overline{X}_1, \dots, \overline{X}_m)$) over k we conclude that L is in fact a finite field extension over k . $\square$

Theorem 1.1.10. Suppose that k is an algebraically closed field. Let $I \subset R = k[X_1, \dots, X_m]$ be an ideal, then $V(I) = \emptyset \iff I = k[X_1, \dots, X_m]$.

Proof: If $I \neq R$, then there exist a maximal ideal $\mathfrak{m} \supset I$. So $V(I) \supset V(\mathfrak{m})$ and it's sufficient to prove that $V(\mathfrak{m}) \neq \emptyset$. By theorem (1.1.9), we have that $L = R/\mathfrak{m}$ is a finite k -extension, so we must have $L = k$ because k is algebraically closed. So there exist $a_i \in k, i = 1, \dots, m$ such that $X_i - a_i \in \mathfrak{m}$ and $(X_1 - a_1, \dots, X_m - a_m) \subset \mathfrak{m}$. However $\varphi : R \rightarrow k : F \mapsto F(a_1, \dots, a_m)$ has kernel $\text{Ker}(\varphi) = (X_1 - a_1, \dots, X_m - a_m)$ and since k is a field, $R/\text{Ker}(\varphi)$ must be a field and $\text{Ker}(\varphi)$ is a maximal ideal. Therefore we conclude that $\mathfrak{m} = (X_1 - a_1, \dots, X_m - a_m)$. Finally $V(X_1 - a_1, \dots, X_m - a_m) = \{a_1, \dots, a_m\} \neq \emptyset$. $\square$

Remark 1.1.11. The hypothesis $k = \overline{k}$ in the previous theorem is crucial. For example if $k = \mathbb{R}$, we have $V(x^2 + 1) = \emptyset$.

Corollary 1.1.12. From the proof of the previous theorem, we see that there exist a bijection

$$\begin{aligned} k^n &\longleftrightarrow \left\{ \begin{array}{c} \text{maximal ideals} \\ \mathfrak{m} \subset R \end{array} \right\} \\ (a_1, \dots, a_n) &\longmapsto (X_1 - a_1, \dots, X_n - a_n) \end{aligned} \quad (1.7)$$

Theorem 1.1.13. (Hilbert Nullstellensatz). Let $A \subset k[X_1, \dots, X_n]$ be an ideal, then

$$IV(A) = \sqrt{A}. \quad (1.8)$$

Proof: The inclusion $\sqrt{A} \subset IV(A)$ is easy. Now, assume $g \in IV(A)$, where $A = (f_1, \dots, f_m)$. This means that we have the hypothesis:

$$f_i(a_1, \dots, a_n) = 0, \forall i = 1, \dots, m \Rightarrow g(a_1, \dots, a_n) = 0 \quad (*) \quad (1.9)$$

Now, define the ideal

$$I = Ak[X_1, \dots, X_{n+1}] + (1 - g(X_1, \dots, X_n)X_{n+1}). \quad (1.10)$$

If I is a proper ideal, then by corollary (1.1.12), we have $I \subset (X_1 - a_1, \dots, X_{n+1} - a_{n+1})$, which is the kernel of the evaluation map $k[X_1, \dots, X_{n+1}] \rightarrow k : f \mapsto f(a_1, \dots, a_{n+1})$. This means that

$$\begin{cases} f_i(a_1, \dots, a_n) = 0, \forall i = 1, \dots, m \\ g(a_1, \dots, a_n)a_{n+1} = 1 \end{cases} \quad (1.11)$$

which contradicts our initial hypothesis (*). Therefore, I is not a proper ideal, so $1 \in I$ and there exists $h_1, \dots, h_{m+1} \in k[X_1, \dots, X_{n+1}]$ such that

$$1 = \sum_{j=1}^m h_j(X_1, \dots, X_{n+1}) f_j(X_1, \dots, X_n) + h_{m+1}(X_1, \dots, X_{n+1})(1 - g(X_1, \dots, X_n)X_{n+1}). \quad (1.12)$$

Now, by projecting this equation onto $k[X_1, \dots, X_{n+1}]/(1 - gX_{n+1})$, we get

$$g^\ell(X_1, \dots, X_n) = \sum_{j=1}^m h_j(X_1, \dots, X_{n+1}) g^\ell \times f_j(X_1, \dots, X_n) \quad (1.13)$$

modulo $(1 - gX_{n+1})$, where $H_j = h_j(X_1, \dots, X_{n+1}) g^\ell \in k[X_1, \dots, X_n]$, $\forall j$ if $\ell \gg 0$. Since $\{\bar{X}_1, \dots, \bar{X}_{n+1}\}$ is algebraically free in $k[X_1, \dots, X_{n+1}]/(1 - gX_{n+1})$, we conclude that, in fact, we have equality

$$g^\ell(X_1, \dots, X_n) = \sum_{j=1}^m H_j(X_1, \dots, X_n) \times f_j(X_1, \dots, X_n) \quad (1.14)$$

in $k[X_1, \dots, X_n]$. □

Remark 1.1.14. We could have used the homomorphism

$$k[X_1, \dots, X_{n+1}] \rightarrow k(X_1, \dots, X_n) : X_{n+1} \mapsto g, \quad X_i \mapsto X_i, \quad i \neq n+1 \quad (1.15)$$

at the end of the previous proof.

1.2 Lecture 3

Definition 1.2.1. We define an affine algebraic set $V \subset k^n$ to be a set of the form $V = V(S)$ for some $S \subset k[\underline{X}]$.

Notation 1.2.2. Let V be an affine algebraic set. We write

$$D_V(g) = \{p \in V : g(p) \neq 0\} = V \setminus V(g). \quad (1.16)$$

These open sets are called “principal open subsets”.

Lemma 1.2.3. Any open subset $U \subset V$ of an affine algebraic set is quasicompact.

Proof: Consider a covering $U = \bigcup_{\lambda \in \Lambda} U_\lambda$. Observe that $V \setminus U = V(g_1, \dots, g_m)$, so we have the equality $U = \bigcup_{i=1}^m D_V(g_i)$. In particular, we see that principal open subsets form a basis for the Zariski topology on V . Therefore, it's sufficient to prove that a principal

open subset is quasicompact. So we assume $U = D_V(g)$. Furthermore, we have $U_\lambda = \bigcup_{i=1}^{n_\lambda} D_V(g_{\lambda,i})$ and

$$U = \bigcup_{\lambda \in \Lambda} \bigcup_{i=1}^{n_\lambda} D_V(g_{\lambda,i}). \quad (1.17)$$

So, without loss of generalization, we can assume $U = \bigcup_{\lambda \in \Lambda} D_V(g_\lambda)$ from beginning. Now

$$\begin{aligned} D_V(g) &= \bigcup_{\lambda \in \Lambda} D_V(g_\lambda) \iff \\ V \cap V(g) &= V \cap V(\langle g_\lambda : \lambda \in \Lambda \rangle) \iff \\ V \cap V(g) &= V \cap V(\langle g_\lambda : \lambda \in I \rangle) \quad (I \subset \Lambda \text{ finite}). \end{aligned} \quad (1.18)$$

In the last equation we used Hilbert basis theorem: The sequence

$$\langle g_{\lambda_1} \rangle \subset \langle g_{\lambda_1}, g_{\lambda_2} \rangle \subset \dots \quad (1.19)$$

must terminate at some point, because $k[\underline{X}]$ is Noetherian. Then we conclude that $D_V(g) = \bigcup_{\lambda \in I} D_V(g_\lambda)$. $\square$

Definition 1.2.4. (Regular functions). Let $V \subset k^n$ be an affine algebraic set and $U \subset V$ be an open set. A map $\varphi : U \rightarrow k$ is said to be *regular* if for any $p \in U$, there exist $p \in U' \subset U$, polynomials $f, g \in k[\underline{X}]$ such that $g(q) \neq 0, \forall q \in U'$ and $\varphi|_{U'} = f/g|_{U'}$.

Lemma 1.2.5. Let $V \subset k^n$ be an affine algebraic set and $g \in k[X_1, \dots, X_n]$. Then the set of regular functions $\varphi : D_V(g) \rightarrow k$ is of the form f/g^m for some $f \in k[X_1, \dots, X_n]$ and $m > 0$. So

$$\mathcal{F}(D_V(g)) \cong k[X_1, \dots, X_n]_g \quad (1.20)$$

where $\mathcal{F}$ denotes the sheaf of regular functions of V .

Proof: First we have $D_V(g) = \bigcup_{i=1}^\ell D_V(g_i)$ and since φ is regular, there exists $f_i, h_i \in k[\underline{X}]$, $i = 1, \dots, \ell$ such that

$$q \in D_V(g_i) \Rightarrow h_i(q) \neq 0 \text{ (so } q \in D_V(h_i)), \varphi|_{D_V(g_i)} = \frac{f_i}{h_i} \Big|_{D_V(g_i)}. \quad (1.21)$$

Now, we have two fundamental relations:

$$\begin{cases} D_V(g_i) \subset D_V(h_i), \forall i = 1, \dots, \ell \\ D_V(g) = \bigcup_{i=1}^\ell D_V(g_i) \end{cases} \quad (1.22)$$

Using $V = V(F_1, \dots, F_m)$ and Hilbert Nullstellensatz (1.1.13), we get

$$\begin{cases} \sqrt{(F_1, \dots, F_m, g_i)} \subset \sqrt{(F_1, \dots, F_m, h_i)}, \forall i = 1, \dots, \ell & (*) \\ \sqrt{(F_1, \dots, F_m, g)} = \sqrt{(F_1, \dots, F_m, g_1, \dots, g_\ell)} & (**) \end{cases} \quad (1.23)$$

From (*), we deduce that $g_i^{n_i} = \sum_j A_j F_j + B_i h_i$ and restricting to V we get $g_i^{n_i} = B_i h_i$. Now, since $\varphi|_{D_V(g_i)} = \frac{B_i f_i}{g_i^{n_i}}$ and $D_V(g_i) = D_V(g_i^{n_i})$, we can **change notation** $B_i f_i \rightarrow f_i$ and $g_i^{n_i} \rightarrow g_i$.

Now we do a trick: since $g_j f_i - f_j g_i = 0$ on $D_V(g_i) \cap D_V(g_j) = D_V(g_i g_j)$, we get

$$g_i g_j (g_j f_i - f_j g_i) = 0 \text{ on } V. \quad (1.24)$$

Furthermore, we still have $\varphi|_{D_V(g_i)} = g_i f_i / g_i^2$. So we **change notation** again: $g_i^2 \rightarrow g_i$ and $g_i f_i \rightarrow f_i$, so we have

$$g_i f_j - g_j f_i = 0, \forall i, j = 1, \dots, \ell \quad (***). \quad (1.25)$$

Now, by equation (**), there exist $m > 0$ such that

$$g^m = \sum_j A_j F_j + \sum_k B_k g_k \quad (1.26)$$

and we get on V :

$$g^m = \sum_k B_k g_k. \quad (1.27)$$

Now, by using equation (***), we have

$$f_i g^m = \sum_k B_k f_i g_k = \sum_k B_k f_k g_i = \left(\sum_k B_k f_k \right) g_i. \quad (1.28)$$

So if $q \in D_V(g_i)$, we have $\varphi(q) = \frac{f_i}{g_i}(q) = \frac{f}{g^m}(q)$, where we set $f = \sum_k B_k f_k$. $\square$

Corollary 1.2.6. Since $V = D_V(1)$, we have $\mathcal{F}(V) = k[X_1, \dots, X_n]$. So the set of regular functions is the set of polynomials on V .

Chapter 2

Some notes from Humphreys

2.1 Pre - varieties and varieties

2.1.1 Projective varieties

Definition 2.1.1. Let V be a vector space, we define its associated projective variety by

$$\mathbb{P}(V) = (V \setminus 0) / \sim \quad (2.1)$$

where we let $v \sim w \iff v = \lambda w$ for some $\lambda \in k^\times$. Suppose that $\dim(V) = n + 1$, then by choosing a basis of V , we identify $V = k^{n+1}$ and we write $\mathbb{P}(V) = \mathbb{P}^n$.

We define the Zariski topology on $\mathbb{P}$ by letting a closed set to be a locus of a homogeneous polynomial. So $X \subset \mathbb{P}^n$ is closed if and only if $X = V(S)$ for some collection $S \subset k[X_0, \dots, X_n]^h$ of homogeneous polynomials. We have a set of isomorphisms ($i = 0, \dots, n$):

$$\phi_i : k[X_1, \dots, X_n] \rightarrow k[X_0, \dots, X_n]^h : g \mapsto X_i^{\deg(g)} g\left(\frac{X_0}{X_i}, \dots, \frac{X_{i-1}}{X_i}, \frac{X_{i+1}}{X_i}, \dots, \frac{X_n}{X_i}\right). \quad (2.2)$$

The inverse of this map takes $\phi_i : h(X_0, \dots, X_n) \mapsto h|_{X_i=1}$. In fact if $h = \sum_{\alpha \in \mathbb{Z}_{\geq 0}^{n+1}} a_\alpha X^\alpha \in k[X_0, \dots, X_n]$ is homogeneous of degree d , then

$$\begin{aligned} \psi_i \phi_i(h) &= X_i^d h\left(\frac{X_0}{X_i}, \dots, \frac{X_{i-1}}{X_i}, 1, \frac{X_{i+1}}{X_i}, \dots, \frac{X_n}{X_i}\right) \\ &= \sum_{\alpha} a_{\alpha} X_1^{\alpha_1} \dots X_{i-1}^{\alpha_{i-1}} X_i^{d - \sum_{j \neq i} \alpha_j} X_{i+1}^{\alpha_{i+1}} \dots X_n^{\alpha_n} \\ &= h, \text{ (because } \sum_j \alpha_j = d). \end{aligned} \quad (2.3)$$

Conversely

$$\phi_i \psi_i(g) = X_i^d g\left(\frac{X_0}{X_i}, \dots, \frac{X_n}{X_i}\right) \Big|_{X_i=1} = g. \quad (2.4)$$

A homogeneous coordinate of $\mathbb{P}^n$ is a map

$$f_i : U_i = \{[x_0, \dots, x_n] : x_i \neq 0\} \rightarrow k^n : [x_0, \dots, x_n] \mapsto \left(\frac{x_0}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_n}{x_i}\right). \quad (2.5)$$

Since $U_i = \mathbb{P}^n \setminus V(x_i)$, it's an open set by definition. Furthermore, the map f_i is an isomorphism. In fact, let $Y \subset U_i$ be a closed set, then $\bar{Y}$ is closed in $\mathbb{P}^n$, so there exist $S \subset k[X_0, \dots, X_n]^h$ such that $\bar{Y} = V(S)$. So we claim that $f_i(Y) = V(\psi_i(S))$. In fact, we have

$$\begin{aligned} x = (x_0, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \in V(\psi_i(S)) &\iff \phi_i(h)(x) = 0 \ \forall h \in S \\ &\iff h[x_1, \dots, x_{i-1}, 1, x_{i+1}, \dots, x_n] = 0 \ \forall h \in S \\ &\iff hf_i^{-1}(x) = 0 \ \forall h \in S \\ &\iff f_i^{-1}(x) \in V(S) \cap U_i = Y \\ &\iff x \in f_i(Y). \end{aligned} \quad (2.6)$$

Conversely, let $Z = V(T) \subset k^n$. We want to show that $f_i^{-1}(Z) \subset U_i$ is closed. We claim that $f_i^{-1}(Z) = V(\phi_i(T)) \cap U_i$, which can be proved in the same lines as before.

Lemma 2.1.2. (Affine characterization for closedness). A subset $X \subset \mathbb{P}^n$ is closed if and only if $X \cap U_i$ is closed in U_i for all $i = 0, \dots, n$.

Lemma 2.1.3. Consider the map

$$\begin{aligned} \varphi : \mathbb{P}^m \times \mathbb{P}^n &\longrightarrow \mathbb{P}^{(m+1)(n+1)-1} \\ ([x_0, \dots, x_m], [y_0, \dots, y_n]) &\longmapsto [\dots, x_i y_j, \dots] \end{aligned} \quad (2.7)$$

where we use $[\dots, x_i y_j, \dots]$ to abbreviate $[x_0 y_0, \dots, x_0 y_n, \dots, x_m y_0, \dots, x_m y_n]$. We claim that $\varphi(\mathbb{P}^m \times \mathbb{P}^n) \subset \mathbb{P}^{(m+1)(n+1)-1}$ is a closed embedding, so it's an isomorphism onto its image.

Proof: Use U_i, V_j, W_{kl} to denote the affine open charts of $\mathbb{P}^m, \mathbb{P}^n$ and $\mathbb{P}^{(m+1)(n+1)-1}$ respectively. The strategy is to show that $\varphi(U_i \times V_j)$ is closed in W_{ij} . However, if $(x, y) \in U_i \times V_j$, then $\varphi(x, y) = [\dots, x_k y_l, \dots]$, where we assume $x_i = 1, y_j = 1$. Now, we observe that the image is of the form $[\dots, u_{kl}, \dots]$, where $u_{kl} = x_k y_l$. Then we conclude that $\varphi(U_i \times V_j)$ is the locus

$$f_{ij}^{-1}(\{[u_{kl} : (k, l) \neq (i, j), u_{kl} - u_{kj} u_{il} = 0, \forall k, l]\}) \stackrel{\text{closed}}{\subset} W_{ij}. \quad (2.8)$$

It remains to show that φ is injective. To do this, it's sufficient to see that $\varphi|_{U_i \times V_j}$ (viewed as a map to $\varphi(U_i \times V_j)$) can be inverted. The equation is just

$$[\dots, u_{kl}, \dots] \mapsto ([u_{0j}, \dots, u_{mj}], [u_{i0}, \dots, u_{in}]). \quad (2.9)$$

□

Corollary 2.1.4. If $X \subset \mathbb{P}^m, Y \subset \mathbb{P}^n$ are closed subsets, then $\varphi(X \times Y) \subset \mathbb{P}^{(m+1)(n+1)-1}$ is a closed subset.

2.1.2 Varieties

Chapter 3

Algebraic Groups

3.1 Basic definitions

Some important observations:

- If $\mathbb{F}$ is a finite field, the map which takes a polynomial in $\mathbb{F}[X_1, \dots, X_n]$ and send to a function $\mathbb{F}^n \rightarrow \mathbb{F}$ is not an isomorphism. For example, if $\mathbb{F}$ has q elements, then $\prod_{q \in \mathbb{F}} (X - q) \in \mathbb{F}[X] \mapsto 0$. However, if $\mathbb{F}$ is infinity and $a_n X^n + a_{n-1} X^{n-1} + \dots + a_0 = 0$, then we can select distinct elements $\alpha_0, \dots, \alpha_n$ and we conclude that

$$\begin{pmatrix} a_0 & \dots & a_n \end{pmatrix} \begin{pmatrix} 1 & 1 & \dots & 1 \\ \alpha_0 & \alpha_1 & \dots & \alpha_n \\ \vdots & \vdots & \dots & \vdots \\ \alpha_0^n & \alpha_1^n & \dots & \alpha_n^n \end{pmatrix} = 0. \quad (3.1)$$

However $\det(\alpha_i^j)_{i,j=0}^n = \prod_{i < j} (\alpha_i - \alpha_j) \neq 0$ and we conclude that $a_0, \dots, a_n = 0$. Now, if $f \in \mathbb{F}[X_1, \dots, X_n]$, then we can write (assuming X_1 appears in f):

$$f = g_d(X_2, \dots, X_n)X_1^d + \dots + g_1(X_2, \dots, X_n)X_1 + g_0(X_2, \dots, X_n). \quad (3.2)$$

So the equality $f = 0$, the previous step for the case $\mathbb{F}[X]$ and an induction on the number of variables yields $g_d, \dots, g_0 = 0$. Then we see that if $|\mathbb{F}| = \infty$, we can identify the ring $\mathbb{F}[X_1, \dots, X_n]$ with polynomial functions $\mathbb{F}^n \rightarrow \mathbb{F}$.

As a consequence of this story, we see that if $|\mathbb{F}| = \infty$, then

$$I(\mathbb{F}^n) = (0) \subset \mathbb{F}[X_1, \dots, X_n]. \quad (3.3)$$

- (Hilbert basis theorem). The Hilbert basis theorem states that if A is noetherian, then $A[X]$ is noetherian. Since $\mathbb{F}[X]$ is a PID (and in consequence noetherian), we conclude by an induction on the number of variables that $\mathbb{F}[X_1, \dots, X_n]$ is noetherian.

Remark 3.1.1. A ring is said to be noetherian if it satisfy the ascending chain condition, or equivalently, if all of its ideals are finitely generated.

- (Algebraic sets). Let $S \subset \mathbb{F}[X_1, \dots, X_n]$, then $V(S) = \{a \in \mathbb{F}^n : f(a) = 0\}$. It's immediate that $V(S) = V(\langle S \rangle)$. The relation $V(\bigcup_{i \in I} S_i) = \bigcap_{i \in I} V(S_i)$ allow us to define the Zariski topology on $\mathbb{F}^n$. Moreover, for any subset $V \subset \mathbb{F}^n$, we let $I(V) = \{f \in \mathbb{F}[X_1, \dots, X_n] : f(a) = 0, \forall a \in V\}$. Then we have

$$\begin{cases} VI(W) = \overline{W} \\ IV(\mathfrak{a}) = \sqrt{\mathfrak{a}} \end{cases} \quad (3.4)$$

We also have the important relation

$$V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a}) \cup V(\mathfrak{b}). \quad (3.5)$$

In fact

$$V(\mathfrak{a}) \cup V(\mathfrak{b}) \underset{\mathfrak{a} \cap \mathfrak{b} \subset \mathfrak{a}, \mathfrak{b}}{\subset} V(\mathfrak{a} \cap \mathfrak{b}) \underset{\mathfrak{a}\mathfrak{b} \subset \mathfrak{a} \cap \mathfrak{b}}{\subset} V(\mathfrak{a}\mathfrak{b}) = V(\{fg : f \in \mathfrak{a}, g \in \mathfrak{b}\}) \subset V(\mathfrak{a}) \cup V(\mathfrak{b}). \quad (3.6)$$

- (Coordinate rings). Let $\mathfrak{a} = I(V)$ where V is an algebraic set. We define the coordinate ring $A[V] = \mathbb{F}[X_1, \dots, X_n]/\mathfrak{a}$.
- (LEX order). We define an ordering on $\mathbb{Z}_{\geq 0}^n$ by $\alpha > \beta$ if for the smallest i such that $\alpha_i \neq \beta_i$ we have $\alpha_i > \beta_i$. So, with a fixed LEX on $\mathbb{F}[X_1, \dots, X_n]$, we can define the leading term of a polynomial: $\text{LT}(f) = a_\alpha X^\alpha$, where α is the smallest tuple, relative to LEX such that $a_\alpha \neq 0$. In the case $\text{LT}(f) = a_\alpha X^\alpha$, we also write $\text{LM}(f) = X^\alpha$.
- (Groebner Basis). Let $\mathfrak{a}$ be an ideal in $\mathbb{F}[X_1, \dots, X_n]$. A groebner basis of $\mathfrak{a}$ is a finite subset $G \subset \mathfrak{a}$ such that $(\text{LT}(g) : g \in G) = \text{LT}(\mathfrak{a})$.

Example 3.1.2. Let $R = \mathbb{F}[X_1, \dots, X_n, Y_1, \dots, Y_n]$ with LEX: $X_1 > \dots > X_n > Y_1 > \dots > Y_n$ and define an ideal $\mathfrak{a} = (X_1 - g_1, \dots, X_n - g_n)$ for $g_i \in \mathbb{F}[Y_1, \dots, Y_n]$. Since $\text{LT}(X_i - g_i) = X_i$, it's clear that $(X_1, \dots, X_n) \subset \text{LT}(\mathfrak{a})$. Now, if $f \in R$ and $X_j \nmid \text{LT}(f)$, $\forall j$, then $f \in \mathbb{F}[Y_1, \dots, Y_n]$ because its smallest term has power $\alpha \in \{0\} \times \mathbb{Z}^n$. So, we need to show that $I \cap \mathbb{F}[Y_1, \dots, Y_n] = (0)$. This can be seen by proving that the kernel of $\varphi : R \rightarrow \mathbb{F}[Y] : X_i \mapsto g_i, Y_i \mapsto Y_i$ is I . The only nontrivial inclusion is $\text{Ker}(\varphi) \subset I$. To see this use a division algorithm to write

$$f = h_1(X_1 - g_1) + \dots + h_n(X_n - g_n) + r \quad (3.7)$$

where $r \neq 0$ implies that no term of r is divisible by $\text{LT}(X_i - g_i) = X_i$, $\forall i$. By applying φ on both sides of this equation we have the desired inclusion. From this we see that $G = (X_1 - g_1, \dots, X_n - g_n)$ is a groebner basis for I .

- (Hilbert Polynomial).

Definition 3.1.3. Let $I \subset \mathbb{F}[X_1, \dots, X_n]$ be an ideal. We define the affine Hilbert function:

$${}^a\text{HF}_I : \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0} : s \mapsto \dim_{\mathbb{F}} \left(\frac{\mathbb{F}[X_1, \dots, X_n]_{\leq s}}{I_{\leq s}} \right). \quad (3.8)$$

Example 3.1.4. For example, the number ${}^a\text{HF}_{(0)}(s)$ is the number of n -tuples of nonnegative integers whose sum is $\leq s$. To compute this, write the generating function

$$\prod_{i=1}^n \frac{1}{1 - X_i} = \sum_{\alpha \in \mathbb{Z}_{\geq 0}^n} X^\alpha. \quad (3.9)$$

Now put $X_i = t$, $\forall i$. We see that

$$(1 - t)^{-n} = \sum_{s \geq 0} \binom{-n}{s} (-1)^s t^s = \sum_{s \geq 0} \binom{n+s}{s} t^s. \quad (3.10)$$

So ${}^a\text{HF}_{(0)}(s) = \binom{n+s}{s} = \frac{1}{n!} s^n + \dots$.

Example 3.1.5. Let $f \in \mathbb{F}[X_1, \dots, X_n]$ be a degree d polynomial. Then

$${}^a\text{HF}_{(f)}(s) = \binom{s+n}{s} - \binom{s-d+n}{s-d} = \frac{d}{(n-1)!} s^{n-1} + \dots. \quad (3.11)$$

An important property of the affine Hilbert function is the following:

Theorem 3.1.6. Let $I \subset \mathbb{F}[X_1, \dots, X_n]$ be an ideal, then

$${}^a\text{HF}_I(s) = {}^a\text{HF}_{\text{LT}(I)}(s), \quad s \geq 0. \quad (3.12)$$

Another main result is the existence of the Hilbert polynomial.

Theorem 3.1.7. There exist a polynomial ${}^a\text{HP}_I(t) \in \mathbb{Q}[t]$ such that

$${}^a\text{HP}_I(s) = {}^a\text{HF}_I(s), \quad \forall s \geq 0. \quad (3.13)$$

Proof (sketch):

- Find a Groebner basis G of I
- Let $M \subset \mathbb{Z}_{\geq 0}^n$ be such that $\{X^\beta\}_{\beta \in M} = \{\text{LM}(g) : g \in G\}$
- Write

$$C(I) = \bigcap_{\beta \in M} C(\beta), \quad C(\beta) = \{\alpha | X^\beta \nmid X^\alpha\} \quad (3.14)$$

as a union of $C(J, \tau) = \{\alpha | \alpha_j = \tau(j) \forall j \in J\}$. To do this use

$$C(\beta) = \bigcup_{i=1}^n \bigcup_{t_i=0}^{\beta_i-1} C(\{i\}, \tau : i \mapsto t_i). \quad (3.15)$$

- Use the formula $|C(J, \tau)_{\leq s}| = \binom{d+s-|\tau|}{d}$, $d = n - |J|$, $|\tau| = \sum_{j \in J} \tau(j)$ and the inclusion-exclusion principle to compute

$${}^a\text{HP}_I(s) = |C(I)_{\leq s}| = \left| \bigcup_{(J, \tau)} C(J, \tau)_{\leq s} \right|. \quad (3.16)$$

Example 3.1.8. Let $I = (X_1 - X_3^2, X_2 - X_3^3)$ and fix the LEX order with $X_1 > X_2 > X_3$. Then we calculate

- $G = \{X_1 - X_3^2, X_2 - X_3^3\}$
- $M = \{(1, 0, 0), (0, 1, 0)\}$
- We have

$$\begin{aligned} C(I) &= C(\{1\}, \tau : 1 \mapsto 0) \cap C(\{2\}, \tau : 2 \mapsto 0) \\ &= C(\{1, 2\}, \tau : 1 \mapsto 0, 2 \mapsto 0). \end{aligned} \quad (3.17)$$

So we conclude that $|C(I)|_{\leq s} = \binom{1+s-0}{1} = s + 1$.

- (Algebraic dimension). Here we define the dimension of an algebraic set.

Definition 3.1.9. Let $V \subset \mathbb{F}^n$ be an algebraic set. We define its dimension as $\dim(V) = \deg {}^a\text{HP}_{I(V)}(t)$. It has the property that it's the biggest integer $d \geq 1$ such that there exist indexes $1 \leq i_1 < \dots < i_d \leq n$ such that $I(V) \cap \mathbb{F}[X_{i_1}, \dots, X_{i_d}] = (0)$.

Example 3.1.10. Let $H_f = V(f)$, f irreducible, then

$$\begin{aligned} \dim(H_f) &= \deg {}^a\text{HP}_{I(H_f)}(t) \\ &= \deg {}^a\text{HP}_{(f)}(t) \\ &= \deg\left(\frac{\deg(f)}{(n-1)!}t^{n-1} + \dots\right) \\ &= n - 1. \end{aligned} \quad (3.18)$$

The dimension has the expected property: If $W \subset V$ is closed, then $\dim(W) \leq \dim(V)$. In fact

$$\begin{aligned} I(V) \subset I(W) &\Rightarrow \\ I(V)_{\leq s} \subset I(W)_{\leq s} &\Rightarrow \\ {}^a\text{HF}_{I(W)}(s) \leq {}^a\text{HF}_{I(V)}(s), \forall s \geq 0 &\Rightarrow \\ \deg {}^a\text{HF}_{I(W)}(t) \leq \deg {}^a\text{HF}_{I(V)}(t). \end{aligned} \quad (3.19)$$

- (Transcendence degree). Let A be a field which is also a finitely generated $\mathbb{F}$ -algebra. We define the transcendence degree of A over $\mathbb{F}$ (and write $\partial_{\mathbb{F}}(A)$) to be largest size of an algebraically independent subset of A over $\mathbb{F}$.

Example 3.1.11. We have $\partial_{\mathbb{F}}(\mathbb{F}[X_1, \dots, X_n]) = n$. Indeed, let $\{f_1, \dots, f_r\}$ be polynomials and suppose that $r > n$. Write

$$\begin{aligned} W_s &= \{f^\beta : \beta \in \mathbb{Z}_{\geq 0}^r, |\beta| \leq s\}, \quad d = \max\{\deg(f_i) : i = 1, \dots, r\} \\ V_s &= \{X^\alpha : |\alpha| \leq s\} \end{aligned} \quad (3.20)$$

Then $W_s \subset V_{ds}$ and

$$|W_s| = \binom{r+s}{s} \stackrel{s \gg 0}{\geq} \binom{n+sd}{sd} = \dim(V_{sd}) \geq \dim(W_s). \quad (3.21)$$

So, for $s \gg 0$ there exist $\{a_\beta\}_\beta \subset \mathbb{F}$ such that $\sum_\beta a_\beta f^\beta = 0$ and $\{f_1, \dots, f_r\}$ is algebraically dependent.

Theorem 3.1.12. Let V be an algebraic set and write $I = I(V)$, then

$$\dim(V) = \partial_{\mathbb{F}}(A[V]) = \partial_{\mathbb{F}}\left(\frac{\mathbb{F}[X_1, \dots, X_n]}{I}\right). \quad (3.22)$$

Proof: Let $d = \dim(V)$, so there exist indexes $1 \leq i_1 < \dots < i_d \leq n$ such that $I \cap \mathbb{F}[X_{i_1}, \dots, X_{i_d}] = (0)$. This is equivalent to say that $[X_{i_1}], \dots, [X_{i_d}] \in A[V]$ are algebraically independent over $\mathbb{F}$. Therefore $d \leq \partial_{\mathbb{F}}(A[V])$. Now, suppose that $\varphi_i = [f_i] \in A[V], i = 1, \dots, m$ are algebraically independent over $\mathbb{F}$ and put $N = \max \deg(f_i)$, then the map

$$\mathbb{F}[Y_1, \dots, Y_m]_{\leq s} \mapsto \frac{\mathbb{F}[X_1, \dots, X_n]_{\leq Ns}}{I_{\leq Ns}} : g \mapsto g(\varphi_1, \dots, \varphi_m) \quad (3.23)$$

is injective. Then, by taking dimensions we get

$$\binom{m+s}{s} \leq {}^a\text{HF}_I(s), \quad \forall s \geq 0. \quad (3.24)$$

We know that both expressions are polynomials in s , so by taking degrees we conclude that $m \leq \dim(V) = d$ and $\partial_{\mathbb{F}}(A[V]) \leq d$. $\square$

With small changes in the above proof, we can verify that in the case of $A[V]$ being an integral domain, we have also $\partial_{\mathbb{F}}(K) = \dim(V)$, $K = \text{Frac}(A[V])$.

Example 3.1.13. Let $C = V(X_1 - X_3^2, X_1 - X_3^3)$, then $I(C) = (X_1 - X_3^2, X_2 - X_3^3)$ and

$$\frac{\mathbb{F}[X_1, X_2, X_3]}{I(C)} \cong \mathbb{F}[X_3] \quad (3.25)$$

so $\partial_{\mathbb{F}}(A[C]) = 1$.

Example 3.1.14. The purpose of this example is to show that if $V \subset \mathbb{F}^n, |\mathbb{F}| = \infty$ is a linear space, then $\dim(V) = \dim_{\mathbb{F}}(V)$.

- Let $\text{Ann}(V) = \text{Span}(f_1, \dots, f_m)$. Since V has finite dimension, we have $v \in V \iff f(v) = 0 \forall f \in \text{Ann}(V)$ and $V = V(\{f_1, \dots, f_m\})$.
- Let $f_i = \sum a_{ij} X_j$, where we identify X_j with the j th coordinate functions in $\mathbb{F}^n$. Now, if $A = (a_{ij})$ and apply reduction to put A in the form:

$$\begin{pmatrix} a_{1j_1} & * & 0 & * & \cdots & 0 & \cdots \\ & & a_{2j_2} & * & \cdots & 0 & \cdots \\ & & & & \ddots & \cdots & \vdots \\ & & & & & 0 & \cdots & a_{mj_m} & \cdots \end{pmatrix} \quad (3.26)$$

with $a_{ij_i} = 1$, $j_1 < j_2 < \dots$. Now put LEX with: $X_{j_1} > \dots > X_{j_m} > X_j$ $j \notin \{j_1, \dots, j_m\}$ and write $f_i = X_{j_i} + h_i$ with $h_i \in \mathbb{F}[X_j : j \notin \{j_1, \dots, j_m\}]$. Then we see that $A[V] \cong \mathbb{F}[X_j : j \notin \{j_1, \dots, j_m\}]$ and $(f_1, \dots, f_m)$ is prime. Therefore $I(V) = (f_1, \dots, f_m)$. Finally

$$\dim(V) = \dim_{\mathbb{F}}(A[V]) = \dim_{\mathbb{F}} \mathbb{F}[X_j : j \notin \{j_1, \dots, j_m\}] = \dim_{\mathbb{F}} V. \quad (3.27)$$

3.2 Tangent Spaces

- (Regular maps). Let $V \subset \mathbb{F}^m, W \subset \mathbb{F}^n$ be algebraic sets. Then a map $f : V \rightarrow W$ is said to be regular if there exist polynomials such that $f = (f_1, \dots, f_m)$. Such a map is continuous in the Zariski topology.

Remark 3.2.1. We can identify $A[V]$ with the set of regular functions $V \rightarrow \mathbb{F}$. In this context, we have a contravariant functor:

$$\varphi : V \rightarrow W \rightsquigarrow \varphi^* : A[W] \rightarrow A[V] : [g] \mapsto [g] \circ \varphi \quad (3.28)$$

Example 3.2.2. Let $\varphi : \mathbb{F} \rightarrow C : x \mapsto (x^3, x^2, x)$ and write $C = (X_1 - X_3^2, X_2 - X_3^3)$. Remember the isomorphism $A[C] \rightarrow \mathbb{F}[Y] : [X_1] \mapsto Y^2, [X_2] \mapsto Y^3, [X_3] \mapsto Y$. So we have

$$\varphi^* : A[C] \rightarrow A[\mathbb{F}] : [X_3] \mapsto [X_3] \circ \varphi. \quad (3.29)$$

The first map is $[X_3] : C \rightarrow \mathbb{F} : (x^3, x^2, x) \mapsto x$ and the last map is $\mathbb{F} \rightarrow C : x \mapsto x$. At the algebraic level, the map φ^* is just the isomorphism $\mathbb{F}[Y] \cong \mathbb{F}[X]$.

- (Tangent spaces). Let V be an algebraic set and $V_p = \mathbb{F}$ be the $A[V]$ -module with action $[f]\alpha \equiv f(p)\alpha$. Let $I(V) = (f_1, \dots, f_m)$. We define

$$\begin{aligned} T_p(V) &= \{v \in \mathbb{F}^n : d(f)_p(v) = 0 \forall f \in I(V)\} \\ &= \{v \in \mathbb{F}^n : d(f_i)_p(v) = 0 \forall i\} \\ &= V(\{d(f_i)_p : i = 1, \dots, m\}) \end{aligned} \quad (3.30)$$

Remark 3.2.3. Caution ! The polynomials of f_i here are not the generators of V , but of $I(V)$. For example, if $V = V(f)$, we can conclude that $T_p(V) = V(df_p)$ only if f is irreducible, i.e., if $I(V) = (f)$.

Here $df_p \equiv \sum_{i=1}^n D_i(f)(p)X_i \in \mathbb{F}[X_1, \dots, X_n]$. We have an isomorphism

$$T_p(V) \rightarrow \text{Der}_{\mathbb{F}}(A[V], V_p) : v \mapsto D_v; \quad D_v([f]) = \sum_{i=1}^n [D_i(f)] \cdot v_i \in \mathbb{F}. \quad (3.31)$$

with inverse $D \mapsto (D([X_i]))_{i=1}^n$. Of course $D_v([f]) = df_p(v)$.

Maybe it will be better to change notation of V_p to ${}^p\mathbb{F}$ or something.

Example 3.2.4. Let $C = (f_1, f_2)$, $f_1 = X_1 - X_3^2$, $f_2 = X_2 - X_3^3$ and $p = (x^3, x^2, x) \in C$. We have

$$d(f_1)_p = 1 - 2xX_3, \quad d(f_2)_p = 1 - 3x^2X_3 \quad (3.32)$$

and $T_p(C) = \mathbb{F}\langle (3x^2, 2x, 1) \rangle$.

Example 3.2.5. The determinant is an irreducible polynomial. Indeed, let $\det \in \mathbb{F}[X_{ij} : 1 \leq i, j \leq n]$. Then, if we apply Laplace theorem on the first row, we get

$$\det = A_{11}X_{11} - A_{12}X_{12} + \dots + (-1)^n A_{1n}X_n \quad (3.33)$$

where A_{ij} denotes the determinant of the matrix obtained by excluding the i th row and j th column. By induction on the number of variables, we assume that A_{ij} are irreducible polynomials. Now, suppose that $\det = fg$, where $\deg(f), \deg(g) > 0$. Choose LEX with $X_{ij} > X_{k\ell}$ if $i < k$ or if $i = k$ and $j < \ell$. Then

$$\text{LT}(A_{11})X_{11} = \text{LT}(\det) = \text{LT}(f)\text{LT}(g). \quad (3.34)$$

Assume that $X_{11} | \text{LT}(f)$ (so $\text{LT}(g) < X_{11}$) and write $f = AX_{11} + B$, $\text{LT}(B) < X_{11}$. Then we must have $AgX_{11} = A_{11}X_{11}$ and $A_{11} = Ag$. Since $\deg(g) > 1$ and A_{11} is irreducible, we must have $g = A_{11}$ and $A = 1$. Now we get

$$-A_{12}X_{12} + \dots + (-1)^n A_{1n}X_n = BA_{11} \quad (3.35)$$

and $-\text{LT}(A_{12})X_{12} = \text{LT}(B)\text{LT}(A_{11})$. Since $X_{22} | \text{LT}(A_{11})$, we are led to $X_{22} | \text{LT}(A_{12})$ which is a contradiction because X_{22} doesn't appear in A_{12} by definition.

Now, since $\det$ is an irreducible homogeneous polynomial, the difference $\det - 1$ must be irreducible. Therefore $I(SL_n \mathbb{F}) = (\det - 1)$ and we can compute the tangent space

$T_I(SL_n) = V(d_I(\det))$. We have

$$\begin{aligned}
d_I \det &= \sum_{k,\ell} D_{k\ell}(\det)(I) X_{k\ell} \\
&= \sum_{k,\ell} \sum_{\sigma \in S_n} D_{k\ell} \prod_{i=1}^n X_{i\sigma(i)}(I) X_{k\ell} \\
&= \sum_{k,\ell} \sum_{\sigma \in S_n} \delta_{\ell,\sigma(k)} \prod_{i \neq k} \delta_{i,\sigma(i)} X_{k\ell} \\
&= \sum_{k,\ell} \delta_{k\ell} X_{k\ell} \quad (\delta_{\ell,\sigma(k)} \prod_{i \neq k} \delta_{i,\sigma(i)} \neq 0 \text{ only if } \sigma = 1) \\
&= \sum_k X_{kk}.
\end{aligned} \tag{3.36}$$

Therefore $T_I SL_n(\mathbb{F}) = \mathfrak{sl}_n \mathbb{F}$, the Lie algebra of traceless matrices.

- (Gauss lemma). We state the Gauss lemma.

Lemma 3.2.6. Let R be a UFD ring (think in $\mathbb{F}[X]$), then $R[X]$ is a UFD.

Proof: Let f and $K = \text{Frac}(R)$. Since $K[X]$ is a UFD, we have $f = FG$, $F, G \in K[X]$ with F irreducible and $\deg(G) < \deg(f)$. Since R is a UFD, there exist $r, s \in R$ such that $rF, sG \in R[X]$. Decompose $rs = t_1 \cdots t_k$ into irreducible elements. Then if $F' = rF, G' = sG$, we have

$$[F'][G'] = [rs][f] = 0 \in R/(t_i)[X] \tag{3.37}$$

because $f \in R[X]$. So, since $R/(t_i)[X]$ is a domain, we see that $t_i | F'$ or $t_i | G'$. Then we see that there exist a partition $\{1, \dots, k\} = I \cup J$ such that $F' / \prod_{i \in I} t_i, G' / \prod_{j \in J} t_j \in R[X]$. It's clear that F' still irreducible, then an induction on degree yields the result. $\square$

Corollary 3.2.7. Let $f, g \in R = \mathbb{F}[X_1, \dots, X_n]$ and assume f irreducible with $f \nmid g$, then there exist $F, G \in R$ such that

$$Gf + Fg = h \in \mathbb{F}[X_1, \dots, X_{n-1}] \setminus 0. \tag{3.38}$$

Proof: Let $K = \text{Frac}(\mathbb{F}[X_1, \dots, X_{n-1}])$, then $f, g \in K[X_n]$ and there exist $A, B \in K[X_n]$ such that $Af + Bg = 1$ (by Gauss lemma f still irreducible as an element of $K[X_n]$). Then there exist $h \in \mathbb{F}[X_1, \dots, X_{n-1}]$ such that $hA, hB \in \mathbb{F}[X_1, \dots, X_{n-1}]$. So by writing $G = hA, F = hB$ we are done. $\square$

Notation 3.2.8. Let $A = \mathbb{F}[X_1, \dots, X_n]/I$, where M is a A -module. We write

$$\mathcal{T}_{A,M} = \{(v_1, \dots, v_n) \in M^n : \sum_i [D_i(f)] \cdot v_i, \forall f \in I\}. \tag{3.39}$$

We have the isomorphism $\mathcal{T}_{A,M} \rightarrow \text{Der}_{\mathbb{F}}(A, M) : v \mapsto D_v$ with inverse

$$D \mapsto (D(\overline{X_i}))_{i=1}^n. \quad (3.40)$$

Now consider the following context, let $K = \text{Frac}(A)$ as A -module (A acts by multiplication). Then we have an isomorphism

$$\text{Der}_{\mathbb{F}}(K, K) \rightarrow \text{Der}_{\mathbb{F}}(A, K) : D \mapsto D|_A. \quad (3.41)$$

In particular we also have the isomorphism $\mathcal{T}_{A,K} \cong \text{Der}_{\mathbb{F}}(K, K)$. We observe that this is also a K -vector space isomorphism, it's a bijection compatible with the K -action. From this isomorphism, we see that $\mathcal{T}_{A,K}$ only depends on K , and the fact that $\text{Frac}(A) = K$.

Theorem 3.2.9. Let $A = \mathbb{F}[X_1, \dots, X_n]/I$ be a domain and $K = \text{Frac}(A)$. Then

$$\partial_{\mathbb{F}}(A) = \dim_K \text{Der}_{\mathbb{F}}(K, K). \quad (3.42)$$

Proof: Let $d = \partial_{\mathbb{F}}(A) = \partial_{\mathbb{F}}(K)$, then there exist $z_1, \dots, z_{d+1}$ such that

- $K = \mathbb{F}(z_1, \dots, z_{d+1})$
- $\{z_1, \dots, z_d\}$ is algebraically independent
- z_d is separable over $\mathbb{F}(z_1, \dots, z_d)$

Let $R = \mathbb{F}[z_1, \dots, z_{d+1}] \subset K$ and $f \neq 0$ irreducible such that $f(z_1, \dots, z_{d+1}) = 0$. Then by corollary (3.2.7), we see that $\alpha : K[Y_1, \dots, Y_{d+1}] \rightarrow K : g \mapsto g(z_1, \dots, z_{d+1})$ has kernel (f) . Therefore

$$\begin{aligned} \mathcal{T}_{R,K} &= \{(v_1, \dots, v_{d+1}) \in K^{d+1} : \sum_i [D_i(f)] \cdot v_i = 0\} \\ &= \{(v_1, \dots, v_{d+1}) \in K^{d+1} : \sum_i D_i(f)(z_1, \dots, z_{d+1}) \cdot v_i = 0\}. \end{aligned} \quad (3.43)$$

We see that $D_i(f)(z_1, \dots, z_{d+1}) = 0 \Rightarrow D_i(f) \in (f)$ and $D_i(f) = 0$. Since $f \neq 0$ and $|\mathbb{F}| = \infty$, we must have $D_i(f) \neq 0$ for some i , so $D_i(f)(z_1, \dots, z_{d+1}) \neq 0$ for some i . Then we conclude that

$$\dim_K \text{Der}_{\mathbb{F}}(K, K) = \dim_K \mathcal{T}_{R,K} = d = \partial_{\mathbb{F}}(K) = \partial_{\mathbb{F}}(A). \quad (3.44)$$

□

Corollary 3.2.10. If $A = \mathbb{F}[X_1, \dots, X_n]/(f_1, \dots, f_m)$, then

$$\partial_{\mathbb{F}}(A) = \dim_K \mathcal{T}_{A,K} = n - \text{Rank}_K([D_i(f_j)]). \quad (3.45)$$

Example 3.2.11. Let $C = V(f_1, f_2)$, $f_1 = X_1 - X_3^2$, $f_2 = X_2 - X_3^3$. We have

$$A[C] = \frac{\mathbb{F}[X_1, X_2, X_3]}{(f_1, f_2)} \rightarrow \mathbb{F}[X_3] : [g] \mapsto g(X_3^3, X_3^2, X_3). \quad (3.46)$$

So $A[C]$ is a domain and we can define $K = \text{Frac}(A[C])$. We compute

$$\text{Rank}_K([D_i(f_j)]) = \text{Rank}_K \begin{pmatrix} 1 & 0 & -2[X_3] \\ 0 & 1 & -3[X_3] \end{pmatrix} = 2. \quad (3.47)$$

Then we conclude $\dim(V) = 3 - 2 = 1$.

- (differential of a regular map). Here we define the notion of differential of a regular map.

Definition 3.2.12. Let $\varphi : V \rightarrow W$ be a regular map between algebraic varieties. Also, let $p \in V, q = \varphi(p) \in W$. We define:

$$d\varphi_p : \text{Der}_{\mathbb{F}}(A[V], V_p) \rightarrow \text{Der}_{\mathbb{F}}(A[W], W_q) : D_v \mapsto D_v \circ \varphi^* \quad (3.48)$$

At the level of tangent spaces, we have:

$$\begin{array}{ccc} \text{Der}_{\mathbb{F}}(A[V], V_p) & \xrightarrow{d\varphi_p} & \text{Der}_{\mathbb{F}}(A[W], W_q) \\ \uparrow & & \downarrow \\ T_p(V) & \xrightarrow{\quad\quad\quad} & T_q(W) \\ & \uparrow \text{red} & \downarrow \text{red} \\ & v & (D_v \circ \varphi^*([Y_i]))_{i=1}^m \end{array}$$

(Note: Red arrows indicate the mapping from the tangent space level to the differential level.)

In more details, if we write $\varphi = (f_1, \dots, f_m)$, then $\varphi^*([Y_i]) = [f_i]$ and we get

$$D_v([f_i]) = \sum_{j=1}^m [D_j(f_i)] \cdot v_j = \sum_{j=1}^m D_j(f_i)(p) v_j. \quad (3.49)$$

Therefore

$$d\varphi_p(v) = (D_j(f_i)(p))_{i,j} \cdot v = J_p(\varphi) \cdot v. \quad (3.50)$$

Example 3.2.13. Let $G \subset \mathbb{F}^{n^2}$ be an algebraic group and $\mu : G \times G \rightarrow G$ be the product of matrices. We have $G \times G \subset \mathbb{F}^{n^2} \times \mathbb{F}^{n^2} \cong \mathbb{F}^{2n^2}$. Now, identify the first n^2

coordinate functions with X_{ij} and the second n^2 coordinate functions with Y_{ij} . So we have

$$\mu_{ij} = \sum_{\ell} X_{i\ell} Y_{\ell j}. \quad (3.51)$$

Then we have

$$\frac{\partial \mu_{ij}}{\partial X_{rs}}(I) = \frac{\partial \mu_{ij}}{\partial Y_{rs}}(I) = \delta_{ir} \delta_{sj}. \quad (3.52)$$

Therefore

$$d_{(I,I)}\mu(A, B) = \left(\sum_{r,s} \frac{\partial \mu_{ij}}{\partial X_{rs}}(I) A_{rs} + \frac{\partial \mu_{ij}}{\partial Y_{rs}}(I) B_{rs} \right)_{ij} = (A_{ij} + B_{ij}) = A + B. \quad (3.53)$$

TODO:

1. Prove that T_1G is a Lie algebra. Remember the star operator.
2. Compute the Lie algebra of SL_n using the fact that $\det -1$ is an irreducible polynomial.

Bibliography

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